INDIVIDUAL ASSIGNMENT

TEST DOCUMENTATION USING IEEE / ISO STANDARDS

Course: Software Testing and Quality Assurance (STQA)

1. Assignment Overview

This individual assignment requires each student to prepare formal test documentation for a software development project they have previously been involved in (academic, internship, or professional project).

The documentation must strictly follow recognized international standards, specifically:

- IEEE 829 – Test Documentation Standard and/or
- ISO/IEC software quality and testing standards (e.g., ISO/IEC 25010)

The goal is to help students apply theoretical standards to real-world software projects and develop professional-quality testing documents.

2. Project Selection Guidelines

Each student must:

- Select one real software project they participated in
- The project may be:
    - Academic project
    - Internship project
    - Personal or organizational software project
- The project should have clear functionality and requirements

3. Required Test Documentation Structure

Students must prepare the following documents (combined as one report):

1. Test Plan (IEEE 829)

Include:

- Test plan identifier
- Introduction and objectives
- Scope of testing
- Test items
- Features to be tested
- Features not to be tested
- Test approach / strategy
- Entry and exit criteria
- Test environment
- Roles and responsibilities
- Schedule and milestones
- Risks and mitigation strategies

2. Test Design Specification

Include:

- Test conditions
- Test coverage mapping (requirements vs tests)
- Testing techniques used

3. Test Case Specification

Include:

- Test case ID
- Description / objective
- Preconditions
- Test steps
- Test data
- Expected results

(At least **10–15 well-designed test cases** are required)

4. Test Execution Report

Include:

- Executed test cases
- Pass / Fail status
- Defects summary (if any)
- Observations during testing

5. Defect Report (IEEE 829)

Include:

- Defect ID
- Description
- Severity and priority
- Steps to reproduce
- Actual vs expected result
- Status

6. Test Summary Report

Include:

- Testing activities performed
- Overall quality assessment
- Coverage summary
- Limitations of testing
- Recommendations

4. Standards Compliance Requirement

- Students must clearly mention:
  - Which IEEE / ISO standard is followed
  - How the document structure aligns with the standard
- Proper section naming and formatting is required

5. Formatting & Submission Guidelines

- Document length: 10–15 pages
- Format: PDF or DOCX
- Use clear headings and tables
- Include diagrams/screenshots if applicable

6. Assessment Criteria

Criteria	Weight
Standard compliance (IEEE / ISO)	30%
Quality of test plan & cases	25%
Project relevance & clarity	20%
Defect analysis & reporting	15%
Organization & presentation	10%

7. Academic Integrity

- The project must be your own work
- Copying templates without customization is not allowed
- Plagiarism and use of Generative AI tools (ChatGPT, DeepSeek and etc) will result in penalties

8. Expected Learning Outcomes

By completing this assignment, students will be able to:

- Apply IEEE/ISO testing standards in practice
- Produce professional test documentation
- Design effective test cases
- Analyze defects systematically
- Understand real-world testing workflows

Group Assignment

Software Testing Tools – Study & Classroom Demonstration

Course: Software Testing and Quality Assurance (STQA)

1. Assignment Overview

This group assignment is designed to help students understand, practice, and demonstrate software testing tools covered in Chapters 1–5. Each group will study assigned tools, perform a basic hands-on demonstration, and present their findings to the class.

The focus is on:

- Understanding the purpose of each tool
- Identifying where and why the tool is used
- Demonstrating basic functionality (conceptual or practical)
- Developing communication and teamwork skills

2. Group Formation

- Students will be divided into 8 or 9 groups
- Each group will be assigned a category of tools
- Every group must ensure equal participation of members

3. Tool Distribution by Group

The tools are distributed below to ensure balanced workload, similar complexity, and comparable number of tools per group.

Group 1 – Unit & Test Frameworks

- JUnit
- PyUnit
- TestNG
- Robot Framework

Group 2 – Web & GUI Automation Tools

- Selenium
- Cypress
- TestComplete

Group 3 – API & Service Testing Tools

- Postman
- REST Assured
- SoapUI

Group 4 – Performance & Load Testing Tools

- JMeter
- Locust
- k6

Group 5 – Security & Vulnerability Testing Tools

- OWASP ZAP
- Burp Suite
- SQL Map

Group 6 – Penetration & Reconnaissance Tools

- Nmap
- Metasploit
- Nessus

Group 7 – Test Management & Reporting Tools

- TestRail
- Jira
- Zephyr Enterprise

Group 8 – Test Data & Environment Tools

- Mockaroo
- Faker
- Docker

Group 9 – Quality Standards & Automation Platforms

- Testsigma
- ISO/IEC 25010
- IEEE 829
- CMMI

4. Tasks for Each Group

Each group must complete the following tasks for every assigned tool:

1. **Tool Introduction**
    - What problem the tool solves
    - Category of testing it belongs to
2. **Purpose & Usage**
    - Why the tool is used
    - Type of projects where it is applicable
3. **Key Features**
    - Main capabilities
    - Strengths of the tool

4. **Demonstration (Mandatory)**
    - Live demo **or**
    - Recorded demo **or**
    - Step-by-step walkthrough with screenshots
5. **Advantages & Limitations**
6. **Real-World Use Case**
    - Example scenario where the tool is applied

5. Presentation Guidelines

- Presentation duration: 15–20 minutes per group
- Demo time: Minimum 5 minutes
- Presentation format: Slides + live/recorded demo
- All group members must actively participate

6. Submission Requirements

Each group must submit:

- Presentation slides (PDF or PPT) and short written summary (2–3 pages)
- Demo evidence (video/screenshots if applicable)

7. Assessment Criteria

Criteria	Weight
Understanding of tools	30%
Demonstration quality	25%
Presentation clarity	20%
Team participation	15%
Time management	10%

8. Academic Integrity

- Plagiarism Use of Generative AI tools (ChatGPT, DeepSeek and etc) is strictly prohibited
- All sources must be acknowledged
- Demonstrations must be original

9. Expected Learning Outcomes

By completing this assignment, students will be able to:

- Identify and classify software testing tools
- Explain practical usage of testing tools
- Demonstrate basic testing activities
- Work effectively in a team environment
- Communicate technical concepts clearly